GrC-Based Image Matting for Mammogram Mass Segmentation

Project Report: Updated Evaluation Version (MIAS and CBIS-DDSM)

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1. SUMMARY

This report updates the earlier project report. The Granular Computing (GrC) image-matting pipeline was re-evaluated under a stricter protocol: a seed-only control, patient-grouped splits and paired confidence intervals. The earlier report scored the pipeline on its own, with an image-level split and no control.

Finding. The full pipeline does not beat the seed-only control. On CBIS-DDSM (269 test images) the pipeline reaches Dice 0.7511 and the seed-only fill 0.8009, a paired difference of -0.0498 (95% CI -0.0638 to -0.0367). MIAS shows the same direction (-0.0328, CI -0.0443 to -0.0230). Results hold for this implementation under this protocol and are not stated for GrC or image matting in general.

2. METHOD AND SETTING

The five layers keep the architecture of the earlier report: (1) two-scale CLAHE with LIPW/LBPW texture maps at 3x3, 5x5 and 7x7; (2) K-means templates with signed cosine matching (K = 8); (3) 7x7 normalised response pooling; (4) RBF-SVM with Platt probabilities, trained on definite pixels only; (5) hybrid prior (0.9 probability + 0.1 normalised intensity) inside a Matting-Laplacian solve, thresholded at alpha >= 0.5. The SVM replaces the probabilistic SVM of the base paper.

The setting is reference-assisted: the crop and trimap come from the reference mask (mass seeds eroded up to 6 px, background beyond a 10 px dilation). The key control is a seed-only fill, which gives each unknown pixel its relative distance to the two seed sets and uses no image and no classifier. The task is therefore boundary refinement given seeds, not lesion detection.

2.1 Datasets

- **CBIS-DDSM (main):** 1,067 images, 611 patients; patient-grouped split with train 642, validation 156 and test 269 images. Masks are algorithm-generated reference masks, not radiologist masks.
- **MIAS (secondary):** 55 mass images from 53 patients, five-fold patient-grouped cross-validation. Labels are filled circles, so MIAS Dice is not comparable with CBIS-DDSM.

3. RESULTS

3.1 Ablation Study and Controls (Dice)

ARM	CBIS-DDSM (N = 269)	MIAS (N = 55)
Manual 3x3 (ablation baseline)	0.7490	0.9263
Single 5x5	0.7492	0.9252
Six-channel, no hybrid	0.7581	0.9241
Full method (six-channel + hybrid)	0.7511	0.9248
Intensity only	0.7516	0.9259
Control: seed-only fill	**0.8009**	**0.9575**
Reference: trimap filled with 0 / with 1	0.3284 / 0.4527	not run
Reference: probability map only	0.0899	not run

Table 1. Ablation arms and controls. The ablation baseline, the control and the trivial references are three different roles.

The five arms differ by under 0.01 Dice, and the seed-only fill is above all of them on both datasets. A refit of the same pipeline moved Dice by about 0.006, so differences among the arms are inside training variation. The hybrid prior and the texture scales add no measurable value here.

3.2 Per-Class Dice

DATASET	CLASS (N)	FULL METHOD	SEED-ONLY FILL	CENTRE ELLIPSE
CBIS-DDSM	CIRC (89)	0.7346	0.7822	0.5410
	SPIC (88)	0.8021	0.8552	0.6260
	MISC (92)	0.7182	0.7669	0.4728
	All (269)	**0.7511**	**0.8009**	**0.5455**
MIAS	CIRC (22)	0.9335	0.9584	not run
	SPIC (19)	0.9198	0.9565	not run
	MISC (14)	0.9178	0.9575	not run
	All (55)	**0.9248**	**0.9575**	**not run**

Table 2. Per-class Dice (test split for CBIS-DDSM; pooled held-out folds for MIAS).

The full method is below the seed-only fill in every class of both datasets. Per-class counts are small, so classes are not ranked against each other. The centre ellipse is a seed-free control built from the crop location alone.

3.3 Classifier Study (CBIS-DDSM test split)

CLASSIFIER	VAL. SEED AUC	TEST BAND AUC	TEST DICE
SVM, main pipeline	0.562	0.497 [0.484, 0.511]	0.7451
SVM + band rows	0.630	0.485 [0.472, 0.498]	0.7436
Gradient boosting	0.651	0.492 [0.480, 0.504]	0.7343
MLP	0.623	0.496 [0.486, 0.506]	0.7256
CNN + band rows	0.817	0.463 [0.449, 0.480]	0.7568
CNN (ROI pixels only)	0.810	0.496 [0.479, 0.512]	0.7411

Table 3. Unknown-band AUC (0.5 is chance). Control: seed-only fill, Dice 0.8009.

Every classifier, including CNNs on raw pixels, scores about 0.5 in the unknown band, even though the CNNs separate the seed pixels well on validation (0.81 to 0.82). Within-image AUC stays at or below 0.55. Better separation of the seeds did not carry over to the boundary, and every run stays below the seed-only fill in Dice.

3.4 Key Paired Comparisons

DATASET	COMPARISON (FIRST MINUS SECOND)	DICE DIFFERENCE [95% CI]
CBIS-DDSM	Full - seed-only (same grid)	-0.0498 [-0.0638, -0.0367]
	Full - seed-only at 10,000 pixels	-0.0348 [-0.0501, -0.0179]
	Full - seed-only at native resolution	-0.0328 [-0.0480, -0.0160]
	Full - intensity only	-0.0006 [-0.0031, +0.0019]
	Prior + probability - centre ellipse (seed-free)	-0.0017 [-0.0038, +0.0005]
MIAS	Full - seed-only	-0.0328 [-0.0443, -0.0230]

Table 4. Paired per-image Dice differences; 95% patient-level bootstrap, 2,000 resamples.

The gap to the seed-only fill narrows with resolution but stays below zero at every setting, so the conclusion does not depend on the 2,500-pixel grid cap. Widening the unknown band, or replacing mask seeds by a click and a box, also leaves the full method below the fill (differences -0.0466 to -0.1723). In the seed-free test the centre ellipse (Dice 0.5455) is as good as any classifier-assisted method, so the crop, which is cut around the reference mask, carries the location information, not the classifier.

4. DISCUSSIONS

- **Seeds carry the result.** Most of the Dice comes from the mask-derived seeds; image evidence in the unknown band adds nothing measurable beyond a distance-based fill.
- **The boundary classifier is the weak link.** Band AUC near 0.5 holds across SVM, boosting, MLP and CNNs. An outline-versus-edge probe (CBIS-DDSM and DMID) found that reference outlines are not closer to image gradient peaks than shifted copies. This is one possible reason, but it cannot separate loose outlines from soft mass boundaries.
- **Granule ceiling.** With perfect labels the ceiling Dice is 0.866 (4 px superpixels) down to 0.434 (16 px), and superpixels did not beat a plain grid at any size. This concerns the granule design tested here only.
- **Scope.** MIAS values are high because circle labels favour a round fill.

5. LIMITATIONS AND REPORTING NOTES

- The setting is reference-assisted; this is not a lesion-detection result.
- CBIS-DDSM masks are algorithm-generated and MIAS labels are circles, so neither is an independent clinical gold standard.
- The test split was reused for diagnostics after the main configuration was fixed. The seed-free tuning grids were widened once and the first run was discarded.
- Layer 5 is solved on at most 2,500 pixels; the resolution check supports the conclusion, but the seed-free comparisons were not rerun at native resolution.
- Confidence intervals cover sampling of test images, not retraining variation.